

ECE250: Signals & Systems Quiz-4

Date: 17-September-2025

Total Points: 10

Duration: 15 Minutes

Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
 2. Institute Plagiarism policy are strictly applicable.
-

[CO2] [Q1]: Consider an LTI system with input and output related through the equation

$$y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau - 3) d\tau$$

- (a) What is the impulse response $h(t)$ for this system? **[5 Points]**
- (b) Determine the response of the system when the input $x(t)$ is as shown in below Figure. **[5 Points]**

